



ACUMENUS

ZEPHYRUS · HOSPITAL OPERATIONS INTELLIGENCE

Ancillary Services Expansion

Radiology · Pathology & Laboratory · Inpatient Pharmacy — Product & Implementation Plan

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Contents

PART I — THE STRATEGIC CASE

1. Executive Summary.....

2. The Evidence: Ancillary Waiting Is Hospital Waiting.....

3. Competitive Landscape and Whitespace.....

46

PART II — SHARED ARCHITECTURE

4. The Milestone Event Spine.....

5. Ingestion and Integration Strategy.....

6. Demo Data, Design Discipline, and Alerting.....

91011

PART III — DEPARTMENT MODULES

7. Radiology.....

8. Pathology & Laboratory.....

9. Inpatient Pharmacy.....

141720

PART IV — CROSS-CUTTING & DELIVERY

10. The Discharge Readiness Vector.....

11. Predictions, Process Intelligence, and Benchmarks.....

12. Phased Roadmap.....

13. Risks and Open Questions.....

24242526

APPENDICES

Appendix A — Source-of-Truth Milestone Maps.....

Appendix B — Claude Code Implementation Prompt.....

Appendix C — Research Provenance.....

293137

PART I — THE STRATEGIC CASE

1. Executive Summary

Zephyrus today instruments the demand side of hospital flow: Emergency Department arrival-to-disposition, real-time bed demand and capacity, the operating-room schedule, and nursing operations. What it does not yet see — except through the thin summary tiles on the RTDC Ancillary Services page — are the three largest sources of waiting inside the hospital: imaging results, laboratory results, and medication readiness. Nearly every stalled disposition, delayed discharge, and late OR start traces back to one of these three departments, and today Zephyrus can only report that the patient is waiting, not why, not where the order is stuck, and not what would clear it.

This plan adds three department modules — Radiology, Pathology & Laboratory, and Inpatient Pharmacy — in the posture Zephyrus already occupies for its existing domains: read-mostly operational intelligence. Zephyrus does not replace the RIS, the LIS, or the pharmacy system. It ingests their events, normalizes them onto a shared milestone spine, evaluates them against explicit service-level clocks, and joins them to the encounters that the ED, RTDC, and Perioperative modules already track. The product output is a single, answerable question rendered at every altitude of the application:

THE UNIFYING PRODUCT QUESTION

"Where is the order stuck, whose patient is it blocking, and what barrier clears it?" Every screen in all three modules answers this question — the cockpit answers it in counts, the workspaces answer it per order, and the Study altitude answers it in trends and Paretos.

The evidence base assembled for this plan (Section 2) shows these departments are not peripheral conveniences but the dominant hidden levers of throughput: imaging drives a 4.4-fold increase in the odds of a four-hour-plus ED stay; the laboratory stage accounts for roughly 69% of ED test turnaround; three of four discharge-delayed patients in a large survey were waiting on medications. The competitive analysis (Section 3) shows the market has left exactly this ground unoccupied: every incumbent is either device-centric, EHR-centric and retrospective, or siloed inside a single department. Nobody connects ancillary readiness to hospital throughput in real time. That connection is Zephyrus's home turf.

Delivery follows proven house patterns and lands in nine to twelve sprints across five phases, each independently shippable and fully demoable before a single interface is connected. The complete, dependency-ordered implementation backlog — sixty tasks with acceptance criteria — is included as Appendix B in the form of a ready-to-run Claude Code prompt.

2. The Evidence: Ancillary Waiting Is Hospital Waiting

The research behind this plan spanned peer-reviewed throughput studies, CAP Q-Probes benchmarking, ASHP national surveys, Joint Commission safety goals, CMS rulemaking, and the published results of the command-center programs at GE HealthCare, TeleTracking, and Qventus. Four findings anchor the strategic case.

2.1 Imaging is the single largest ancillary driver of ED length of stay

Patients who receive radiology diagnostics are 4.4 times more likely to exceed four-hour ED stays, and a single CT order can add up to 150 minutes of ED waiting. The relationship is causal and monotonic: a 60-minute reduction in radiology turnaround time yields a 35–45 minute reduction in average ED length of stay. On the inpatient side, discharge-aware imaging prioritization is worth 0.27–0.49 bed-days per prioritized order — the published Qventus figures — and GE HealthCare's command-center imaging tile produced a 48% reduction in order-to-completion time for CT and MR at its flagship deployments.

2.2 The lab owns the ED clock, and its errors are flow errors

The laboratory stage accounts for approximately 69% of total ED test turnaround, and each one-minute increase in lab TAT adds roughly 0.32 minutes of ED length of stay. Meanwhile 60–70% of all laboratory errors occur in the pre-analytic phase — collection, labeling, transport — which is to say they are flow problems, not analyzer problems, and are therefore visible to (and addressable by) a flow-analytics product rather than an LIS feature. The morning-lab cycle gates discharge decisions: results that land after physician pre-rounding push discharge decisions to the afternoon or the next day.

2.3 Medication readiness is the silent discharge and boarding tax

Boarded ED patients face a 54% higher risk of home-medication delays and a 149% higher risk of delayed antibiotic administration. In a large UK survey, roughly three in four discharge-delayed patients attributed the delay to waiting for medicines; a US quality-improvement program measured bedside discharge-medication delivery at 4.9 ± 2.6 hours. The sepsis SEP-1 three-hour antibiotic window moved into Medicare pay-for-performance, making the pharmacy's order-to-first-dose telemetry a revenue-relevant clock, not just a quality nicety.

2.4 The staffing context makes instrumentation urgent

All three departments are operating under historic workforce constraints: the radiologist shortage has saturated even the national teleradiology overflow market; medical-laboratory-scientist vacancy rates run as high as 25% with a pipeline meeting barely half of demand; and roughly 88% of hospitals report shortages of experienced pharmacy technicians, rising above 92% for sterile-compounding technicians. Departments that cannot add staff must find capacity in flow — which is precisely what milestone-level instrumentation surfaces.

Department	Headline flow evidence	What it gates
Radiology	4.4× odds of 4h+ ED stay; −60 min TAT → −35–45 min ED LOS; 0.27–0.49 bed-days per prioritized order; command-center tile −48% order → completion	ED disposition · inpatient discharge · IR throughput
Pathology & Lab	Lab stage ≈ 69% of ED test TAT; +1 min TAT ≈ +0.32 min ED LOS; 60–70% of lab errors are pre-analytic; troponin < 60 min gates chest-pain disposition	ED disposition · AM discharge decisions · OR starts (T&S, frozen sections)
Inpatient Pharmacy	149% higher delayed-antibiotic risk for boarders; ~3 of 4 discharge delays blame medication waits; bedside delivery 4.9 ± 2.6 h; SEP-1 now pay-for-performance	Discharge readiness · ED boarding care · sepsis quality dollars

3. Competitive Landscape and Whitespace

The ancillary-analytics market is crowded in every direction except the one that matters to a hospital operations platform. Incumbents cluster into three camps, each structurally unable to make the throughput connection.

3.1 Device- and vendor-centric platforms

BD HealthSight sees Pyxis; Omnicell One sees the XT fleet; Sectra's analytics see the PACS. These products are excellent at their own device telemetry and useless outside it — a BD dashboard cannot tell you that the unverified STAT order belongs to a patient boarding in ED bay 12, because BD does not know the patient is boarding. Their data is a subset of what Zephyrus would ingest; their context is a fraction of what Zephyrus already has.

3.2 EHR-centric, retrospective reporting

Epic's Radar dashboards, SlicerDicer, and the Clarity/Caboodle warehouse are ubiquitous and genuinely useful — for yesterday. The warehouse refreshes nightly; the dashboards are departmental; and the correlation across domains (this pending CT is blocking that discharge, which is blocking that ED admission) is left as an exercise for the analyst. Epic shops supplement with Tableau or Power BI over Clarity, inheriting the same latency and the same siloing.

3.3 Department-siloed analytics specialists

Roche/Viewics and hc1 build credible lab-operations analytics; Bluesight owns diversion monitoring; LogicStream models drug-shortage impact. Each stops at its department's door. LeanTaaS — the closest philosophical competitor in capacity optimization — has no radiology product at all, leaving scanner-level flow optimization as open ground.

3.4 The whitespace, precisely stated

POSITIONING

No product on the market today connects ancillary readiness to hospital throughput in real time. The imaging order that blocks a discharge, the troponin that gates an ED disposition, the discharge prescription that holds a bed — these joins require exactly the data Zephyrus already has (ED tracking, bed requests, discharge priorities, OR schedule, barriers) plus the milestone feeds this plan adds. The cross-module joinery is the moat.

Product / camp	Sees	Blind to	Latency
BD HealthSight / Omnicell One	Own dispensing devices, infusion	Patient context, ED/discharge state, other vendors	Near-real-time
Epic Radar / SlicerDicer / Clarity	Everything in the EHR, per department	Cross-domain flow joins; anything non-Epic	Nightly (mostly)
Viewics · hc1 (lab)	Lab volumes, TAT, QC	What the pending result is blocking	Near-real-time
Bluesight · LogicStream	Diversion; shortage impact	Flow, throughput, encounters	Varies
GE Command Center	Cross-domain tiles (excellent)	Requires GE program + wall	Real-time

		build-out	
LeanTaaS iQueue	OR, infusion, beds	No radiology, lab, or pharmacy product	Predictive
Zephyrus (this plan)	Milestone spine × ED/RTDC/Periop context	—	Real-time + warehouse backfill

3.5 Relationship to existing Zephyrus surfaces

The existing Ancillary Services page under RTDC remains the cross-department consumer view; its per-unit tiles gain drill-through links into the three new department workspaces. The ED tracking board gains per-patient pending-ancillary chips — the pattern a 2024 JACR paper documented as eliminating "where's my read?" phone calls. RTDC Discharge Priorities gains the readiness vector described in Section 10. The Perioperative case timeline gains blood-bank and frozen-section gates. And every ancillary breach can be annotated with a barrier record, feeding the existing root-cause and PDSA improvement workflow. The three modules do not bolt onto Zephyrus; they complete circuits the platform already half-owns.

PART II — SHARED ARCHITECTURE

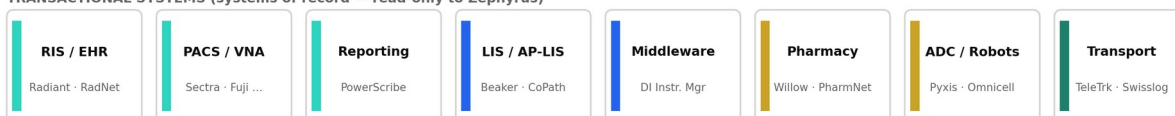
4. The Milestone Event Spine

Strip away the clinical particulars and all three departments reduce to the same operational shape: an ordered thing — an imaging order, a specimen, a medication order — moving through timestamped milestones toward a terminal state, with priority classes, service-level clocks, and barriers along the way. Zephyrus builds this shape once, as a shared spine, and each department contributes a satellite of domain-specific detail. The spine is what makes the cross-module joins cheap and the fourth, fifth, and sixth departments (respiratory therapy, dialysis, rehabilitation) nearly free.

One spine, three departments

Every ancillary feed reduces to the same shape: timestamped milestones on an order, evaluated against explicit SLA clocks.

TRANSACTIONAL SYSTEMS (systems of record — read-only to Zephyrus)



FEEDS (standards-first; no per-vendor connectors)



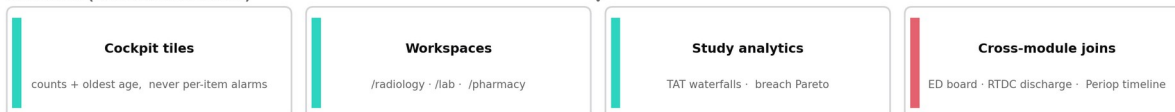
INTEGRATION RUNTIME (already operational — 2026-07-10)



MILESTONE SPINE (new — Phase 0)



SURFACES (role-aware altitudes)



Read-only posture: Zephyrus never writes back to the RIS, LIS, or pharmacy system. Barrier annotation and SLA configuration live in Zephyrus tables only.

Figure 1 — The shared ancillary architecture: transactional systems feed standards-based interfaces into the existing integration runtime; the new milestone spine normalizes everything; role-aware surfaces and cross-module joins consume it.

4.1 The five spine tables

Table	Purpose
prod.ancillary_orders	One row per order / specimen / medication order, discriminated by department; carries encounter linkage, patient class, priority, current state.
prod.ancillary_milestones	Append-only timestamped milestone events with source attribution and provenance linkage. Conflicting assertions from two systems are both retained.

Table	Purpose
prod.ancillary_sla_definitions	Explicit clock definitions: start milestone, stop milestone, warning and breach thresholds, scoped by priority and patient class. Thresholds are data, never code.
prod.ancillary_breaches	Materialized breach records with open/clear lifecycle, linked to the existing Barrier model for root-cause workflow.
hosp_ref.ancillary_milestone_types	Reference catalog per department: code, label, expected sequence, expected source-of-truth system.

4.2 Design decisions that carry the product

Explicit, auditable clock definitions. The radiology benchmarking literature's first finding is that "turnaround time" has no universal definition — order-to-final, exam-complete-to-final, and image-available-to-preliminary are different numbers that get casually interchanged. Every KPI in these modules references a row in the SLA definitions table, and every KPI tile displays its clock definition on hover, using the metric-definition pattern the Ancillary Services page already established. Targets are percentile-based — median and 90th percentile, never mean alone.

Append-only milestones with provenance. Milestones carry the asserting source and link to the integration provenance records. When the RIS says the exam started at 10:02 and the modality's MPPS message says 10:07, both assertions are kept; the SLA engine selects per the source-of-truth catalog (Appendix A). This is what makes the analytics defensible in a room full of skeptical department directors.

Encounter joinery. Every ancillary order links to the encounter Zephyrus already tracks, inheriting patient class, unit, attending — and, critically, membership in the ED tracking board, the boarding list, and the discharge priority queue. This single foreign key is the entire cross-module strategy.

Department specificity lives in satellites. Modality, scanner, and read detail in the radiology tables; specimen, result, and case detail in the lab tables; formulary, verification, and dispensing detail in the pharmacy tables. The spine stays generic; the satellites stay honest to each department's vocabulary.

Process-mining registration from day one. Each department registers OCEL object types and event types mirroring its milestone codes, aligned to the D1 (Lab), D5 (Imaging), and D7 (Pathology) process families the existing catalog already anticipates. The Study altitude's conformance and performance views work on ancillary flows without new UI.

5. Ingestion and Integration Strategy

The integration runtime operationalized on 2026-07-10 — sources with protocol health, raw inbound messages, canonical event records, connector watermarks, dead letters, and replayable ingestion jobs — is the substrate. Each department adds source registrations, pure-function message parsers, a milestone projector, and an SLA evaluator. Parsers are

deliberately boring: a raw inbound message goes in, zero or more canonical events come out, and the whole pipeline is replayable through the existing event-replay jobs.

5.1 Feed strategy by protocol

Protocol	Role	Notes
HL7 v2 (ORM/OML/OMI, ORU, RDE/RDS, SIU, ADT)	Primary real-time	The workhorse. Every hospital has these on an engine. Zephyrus receives via MLLP listener or engine HTTPS push.
DICOM MPPS	Machine-truth exam start/end	Uneven support at older sites; RIS Begin/End Exam events are the documented fallback.
FHIR R4 + Bulk Data	Secondary / backfill	ServiceRequest, ImagingStudy, DiagnosticReport, Specimen, Observation, MedicationRequest/Dispense. US Core does NOT cover MedicationAdministration — BCMA events come from the warehouse.
Warehouse extracts (Clarity / Caboodle)	Retrospective baselining; fields with no interface	Nightly. Feeds the Study altitude and honest-labeled administration metrics, never the real-time tiles.
Vendor events (ADC transactions, tube system, middleware, CTRM, transport)	Segment enrichment	Each optional; every screen defines a degraded mode that renders with coarser milestones when a feed is absent.

5.2 Engine-agnosticism as a requirement

NextGen closed-sourced Mirth Connect from version 4.6 in March 2025 — version 4.5.2 was the last open-source release — spawning the BridgeLink and Open Integration Engine forks and pushing roughly two-thirds of health-IT leaders to re-evaluate their integration strategy. The practical lesson for Zephyrus: speak MLLP and HTTPS, assume nothing about the site's engine (Rhapsody, Cloverleaf, Iguana, a Mirth fork), and let the existing sources/watermarks/replay machinery own reliability.

5.3 The Epic-first posture

Epic holds 43.7% of US acute-care hospitals and 56.9% of beds (2026 KLAS figures on 2025 data), and its ancillary modules — Radiant, Beaker, and Willow — ride that share. Oracle Health continues to shed hospitals; MEDITECH Expanse holds the community segment. An Epic-first integration posture (real-time HL7 v2 plus verify-queue events via open.epic, with Clarity/Caboodle backfill) therefore covers the majority of target beds, with Oracle and MEDITECH as second-tier variants of the same standards-based feeds rather than separate builds.

6. Demo Data, Design Discipline, and Alerting

6.1 Demo-first is a hard requirement

Every screen in all three modules must work in demo mode before any interface exists — this is how the ED, RTDC, and Perioperative modules shipped, and it is what makes the product

demoable to a prospect and testable in CI. The operational demo-data service gains three generators driven by the demo clock and distribution sampler: radiology arrivals with realistic stage distributions (the published ED CT median of 108 minutes with its 57–182 interquartile range is the anchor), night-and-weekend TAT degradation, and portable exams that skip transport; a lab generator with the 03:00–06:00 AM-draw wave, hemolysis-and-recollect loops at ED-realistic rates, auto-verification pass-through, and median seven-minute critical-value callbacks; and a pharmacy generator with the 09:00–11:00 verification-queue surge, STAT and first-dose clock classes, IV-batch cycles with cutoffs, and discharge-medication statuses tied to the demo discharge list. The demo invariant service enforces strict milestone ordering, breach-SLA consistency, and cross-module referential coherence: a demo discharge blocked on imaging must reference a real demo imaging order.

6.2 Token Canon under status pressure

These are the most status-dense screens in the application, which makes them the highest alarm-fatigue risk and the strictest test of the design system. The non-negotiables carry over with no exceptions: the blue/slate healthcare palette governs all operational surfaces; status colors are earned — amber only past a warning threshold, coral only on true breach — and thresholds live in the SLA definitions table, never in component code; status is never conveyed by color alone; every metric renders in tabular numerals; surfaces use the Card/Panel primitives; and wall-density micro-captions are sanctioned only inside the cockpit component directory.

THE GE LESSON, ADOPTED

GE's command-center imaging tile flags a CT exam amber at 18 hours and red at 24 — and pairs the color with a human-actionable barrier field. That restraint (aging thresholds, not per-event alarms) is credited with the tile's 48% order-to-completion improvement. Zephyrus adopts the philosophy: cockpit tiles surface counts and oldest-item age; individual alarms never leave the workspace altitude.

6.3 Role-aware altitudes

The role switcher works here exactly as it does elsewhere in Zephyrus: frontline users see worklists and timers; operations leaders see queues, staffing context, and barrier Paretos; executives see SLA compliance trends, benchmark deltas, and flow impact expressed in bed-days recovered. Same pages, altitude-appropriate default lenses.

PART III — DEPARTMENT MODULES

7. Radiology

Radiology goes first because it has the strongest evidence of throughput impact, the best-standardized feeds, and the clearest command-center precedent. The module instruments the imaging value stream as fourteen timestampable milestones; every KPI is a delta between two of them, sliced by priority, modality, and patient class.

Radiology — the imaging value stream

Fourteen timestampable milestones; every KPI is a delta between two of them, sliced by priority × modality × patient class.

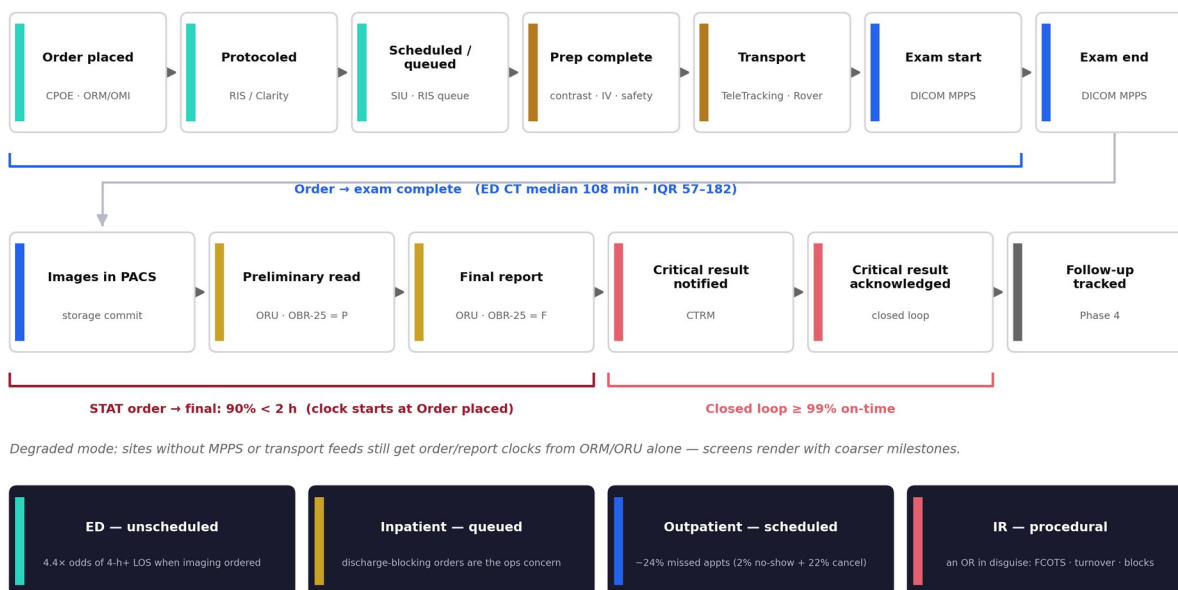


Figure 2 — The imaging value stream. Machine truth for exam start/end comes from DICOM MPPS; report milestones ride the ORU result-status field; the critical-results loop closes through the CTRM.

7.1 How the flow differs by setting and modality

The ED flow is unscheduled and urgency-dominated: preliminary reads matter, portables go to the patient, and imaging is the largest single length-of-stay lever. The inpatient flow is a queue shaped by transport and scanner contention, where the operational question is which orders block today's discharges. The outpatient flow is scheduled, and its levers — slot templates, no-shows (about 2% true no-shows plus 22% cancellations, for roughly 24% missed appointments), and days-to-next-available — are access problems rather than throughput problems. Modalities differ just as sharply: XR is high-volume and prep-free; CT is the ED workhorse with protocoling, contrast screening, and IV access in its critical path (a device-productivity study measured 136 minutes of CT room time per exam against only 60 minutes of scanning); MRI adds mandatory safety screening, with IV and port issues the single most frequent delay; ultrasound is sonographer-limited rather than room-limited; interventional radiology behaves like an operating room and is measured like one; and nuclear medicine is scheduled around radiopharmaceutical decay.

7.2 KPI canon

KPI	Benchmark / target guidance
STAT order → final (CT/MR)	60–120 min; common contract: 90% of STAT reads < 2 h
ED image-available → final	< 30–60 min; stroke/ICH tighter
Routine inpatient / outpatient report TAT	Hours–24 h inpatient; 24–72 h outpatient
ED CT order → exam complete	Published median 108 min (IQR 57–182) — the demo-data anchor
Order → exam start, by patient class	The discharge-blocking metric; sliced by "on today's discharge list"
Transport request → arrival	Lean projects show 20–35% cycle reduction available
Modality utilization % / idle-gap minutes	~75% effective utilization; MRI best practice > 85%
Worklist depth + unread backlog growth	Leading indicator, by subspecialty and shift
Critical results: notified %, closed-loop ack %	Joint Commission NPSG.02.03.01; ≥ 99% on-time achievable
Follow-up recommendation completion	Published completion only 28–77% — a large safety gap
IR: first-case on-time starts, turnover, utilization	Reuse Periop implementations; IR turnover ~20 min median

7.3 Barrier catalog

The module enumerates barriers as first-class data feeding the existing root-cause workflow: transport delay; IV access and contrast screening (creatinine holds); incomplete MRI safety screening; protocoling queues mis-routed by subspecialty; radiologist coverage gaps on nights and weekends — the dominant 2025–26 constraint, with the national teleradiology overflow market itself saturated; scanner downtime; ED-versus-inpatient contention on shared scanners with its systematic outpatient bump list; patients not ready (NPO status, sedation); portable backlog generated by ED boarding; and open results-acknowledgment loops. One regulatory note: the PAMA appropriate-use-criteria program was rescinded — CMS removed all claim edits effective January 2025 with no restart timeline — so imaging-appropriateness analytics are a future differentiator, not a compliance requirement.

7.4 Screens

Route	Surface	What it shows
/radiology	Imaging Flow Board (workspace)	Aging-order heat map by modality × aging bucket; breach chips; barrier drawer; discharge-blocking lens
/radiology/worklist	Order Worklist (workspace)	Filterable orders with per-order milestone timeline; transport segment timers when the feed exists
/radiology/modality	Modality Utilization (workspace)	Per-scanner day view: exam blocks vs idle gaps, ED/IP/OP mix, downtime
/radiology/reads	Reads & Results (workspace)	Worklist depth by subspecialty; prelim – final aging; critical-results closed-loop tracker
/analytics/radiology-tat	TAT Analytics (study)	Segment waterfall; percentile trends;

Route	Surface	What it shows
		night/weekend degradation; breach Pareto by barrier
/analytics/ir-utilization	IR Suite (study)	FCOTS, turnover, utilization — the Periop analytics cloned onto IR rooms
Cockpit tile	Imaging health	Oldest unread by priority; open breach count; scanners down

7.5 Transactional systems and integration posture

System class	Market leaders (2025–26)	Zephyrus integration
EHR / RIS	Epic Radiant (43.7% hospitals / 56.9% beds) · Oracle RadNet (21.9%) · MEDITECH Expanse (14.7%). The standalone RIS has collapsed into the EHR.	ORM/OMI + ORU real-time; Radiant Clarity for protocoling and backfill
PACS / VNA	Sectra (winning ~25% of new large-PACS deals) · Fujifilm · Agfa · Merative · Philips · GE (declining)	High heterogeneity → no per-PACS connectors; MPPS + storage-commitment / ImagingStudy events
Voice / reporting	Nuance PowerScribe (~80% share; 360 sunseting toward PowerScribe One) · Solventum Fluency	Report status arrives via ORU (preliminary/final/corrected); CTRM feed for critical results
Worklist orchestration	PowerScribe Workflow Orchestration (ex-Primordial) · Intelrad InteleOrchestrator (ex-Clario)	Optional worklist-depth feed; otherwise derived from the unread ORU gap
Transport	TeleTracking Capacity IQ · Epic Rover / Transport	Transport request → assign → in-transit → complete milestones

RADIOLOGY SATELLITE TABLES

prod.rad_exams (modality, subspecialty, protocol, contrast, portable, scanner, IR flag) ·
 prod.rad_scanners (status, downtime windows) · prod.rad_reads (prelim/final, radiologist,
 teleradiology flag, addenda) · prod.rad_critical_results (finding class, notified/acked, escalation state)
 · hosp_ref.rad_modalities, rad_subspecialties.

8. Pathology & Laboratory

The lab module covers both the clinical laboratory — chemistry, hematology, coagulation, microbiology, and the blood bank — and anatomic pathology, whose workflow differs structurally enough to warrant its own sub-spine. The clinical-lab spine is unusually well served by standards: the HL7 ORU result message natively carries the collection timestamp (OBR-7), the received-in-lab timestamp (OBR-14), and the result-status transitions, which means the three load-bearing TAT segments — collect-to-receipt, receipt-to-result, and order-to-result — are available at every hospital without any vendor-specific feed.

Pathology & Laboratory — specimen and result flow

Clinical lab rides the pre-analytic → analytic → post-analytic spine; anatomic pathology forks through the overnight histology batch.



Figure 3 — Clinical-lab and anatomic-pathology flows. The pre-analytic phase — where 60–70% of lab errors live — is fully visible to a flow product; the overnight histology batch is the structural TAT floor in anatomic pathology.

8.1 KPI canon

KPI	Benchmark
ED troponin order → result	< 60 min consensus gate (hs-troponin algorithms require it)
STAT TAT compliance	90% within 60 min is the common lab-defined target
Collect → receive (transport segment)	CAP Q-Probes potassium: median 28 min delivery / 47 analytic / 82 total
Specimen rejection rate	Q-Probes median 0.31%; hemolysis is 40–70% of rejections; ED hemolysis 8–29%
Blood culture contamination	≤ 3% accepted threshold; top decile ≤ 1%

KPI	Benchmark
Auto-verification rate	Baseline 40–60%; > 90% achievable; middleware claims up to 95%
Critical value time-to-notify	Q-Probes median 7 min; documentation completeness tracked alongside
AM-draw readiness	% of AM orders resulted before 09:00 rounds — the discharge-decision gate
AP routine biopsy TAT	CAP guidance: ≥ 90% final within 2 days (3 for complex)
Frozen section TAT	90% of single-block frozens < 20 min; timer tied to the live OR case
Blood bank readiness	T&S ~10–15 min to issue vs ~90 min full crossmatch; % OR cases ready pre-start
Micro culture milestones	Gram stain ~1 day · organism ID ~2 · susceptibility ~3 — stewardship events

8.2 Barrier catalog

The enumerated barriers: the phlebotomy staffing wave (AM-draw compression into 03:00–06:00); nurse-collect versus lab-collect misrouting; the hemolysis-and-recollect loop, which silently adds a full TAT cycle; pneumatic tube downtime and mis-routing; accessioning queues; STAT/routine mixing on shared analyzers; the overnight histology batch; analyzer downtime; the medical-laboratory-scientist shortage (vacancies to 25%, pipeline roughly 46% short of demand) and the pathologist shortage (a projected ~4,230-FTE national shortfall by 2037) as staffing context for TAT drift; and courier schedules bounding outreach turnaround.

8.3 Screens

Route	Surface	What it shows
/lab	Lab Flow Board (workspace)	Live stage waterfall (collect → receive → result → verify → notify); STAT-compliance tiles; pre-analytic quality strip (hemolysis, rejections, contamination vs benchmarks)
/lab/specimens	Specimen Tracker (workspace)	Per-specimen milestone timeline; recollect-loop flagging; tube-transport segment when the feed exists
/lab/pending-decisions	Decision-Pending Results (workspace)	THE HERO: pending results joined to what they gate — an ED disposition, a discharge, an OR start — sorted by downstream impact
/lab/blood-bank	Blood Bank Readiness (workspace)	T&S / crossmatch status against today's OR schedule; MTP activity
/lab/anatomic-path	AP Case Aging (workspace)	Case-aging board by stage vs the CAP two-day guidance; frozen-section live timer on the OR case
/analytics/lab-tat	Lab TAT Analytics (study)	Percentile trends by test/priority/shift; AM-readiness curve; auto-verification trend; breach Pareto
Cockpit tile	Lab health	STAT compliance now; oldest decision-pending result; critical-value SLA state

THE DIFFERENTIATING FEATURE

Decision-Pending Results is the plan's single most differentiated lab feature. A decision-class column on the test catalog (ED-disposition / discharge-gate / OR-gate) joins every pending result to what it is blocking. No LIS dashboard can build this view — the LIS does not know the patient is a discharge candidate. Zephyrus does.

8.4 Transactional systems and integration posture

System class	Market leaders (2025–26)	Zephyrus integration
LIS	Epic Beaker (~35% and rising with Epic wins) · Oracle PathNet (~22%) · Clinisys/Sunquest · SCC SoftLab · Orchard · Meditech	OML/ORM + ORU R01 real-time (OBR-7/14/22 timestamps); Beaker Clarity extracts for backfill
AP LIS	Epic Beaker AP · Clinisys CoPathPlus · Cerner PathNet AP	Case-status events; Clarity backfill for stage detail
Middleware	Data Innovations Instrument Manager (6,000+ labs, 1,000+ drivers) · Roche cobas infinity	Auto-verification %, instrument status and downtime, reflex events — the richest optional feed
Blood bank	WellSky HCLL · Haemonetics SafeTrace Tx · Epic Beaker BB · SCC SoftBank	Type & screen / crossmatch / unit-issue events
Specimen transport	Swisslog TransLogic (2,000+ North American installs; RFID chain-of-custody)	Tube-carrier events → the transport segment
Collection	Epic Rover bedside barcode scanning	Collection timestamps arrive inside ORM/ORU (OBR-7); no separate feed needed

LAB SATELLITE TABLES

prod.lab_specimens (type, container, collector role, rejection code, parent-specimen link for recollects) · prod.lab_results (LOINC + local code, status, auto-verified flag, critical flag, analyzer) · prod.lab_critical_values (notified, read-back, minutes-to-notify) · prod.ap_cases (case type, block/slide counts, stage timestamps, OR-case link for frozens) · prod.bb_readiness (T&S / crossmatch status vs OR case) · hosp_ref.lab_test_catalog with the decision_class column.

9. Inpatient Pharmacy

Pharmacy closes the loop because it completes the discharge readiness vector and because its verification queue is the purest real-time signal in the plan: Epic publishes an event every time an inpatient medication order enters or leaves the pharmacist verification queue, making queue-depth telemetry — which no national benchmark even exists for — an immediately available differentiator. The medication loop's speed is decided at its preparation branch: a med stocked in the unit's automated dispensing cabinet moves in minutes; one routed through the IV room inherits USP <797> beyond-use-date batching; one picked centrally inherits delivery logistics.

Inpatient Pharmacy — the medication loop

Order → verify → prepare → dispense → deliver → administer, with priority clocks and the branch structure that decides speed.

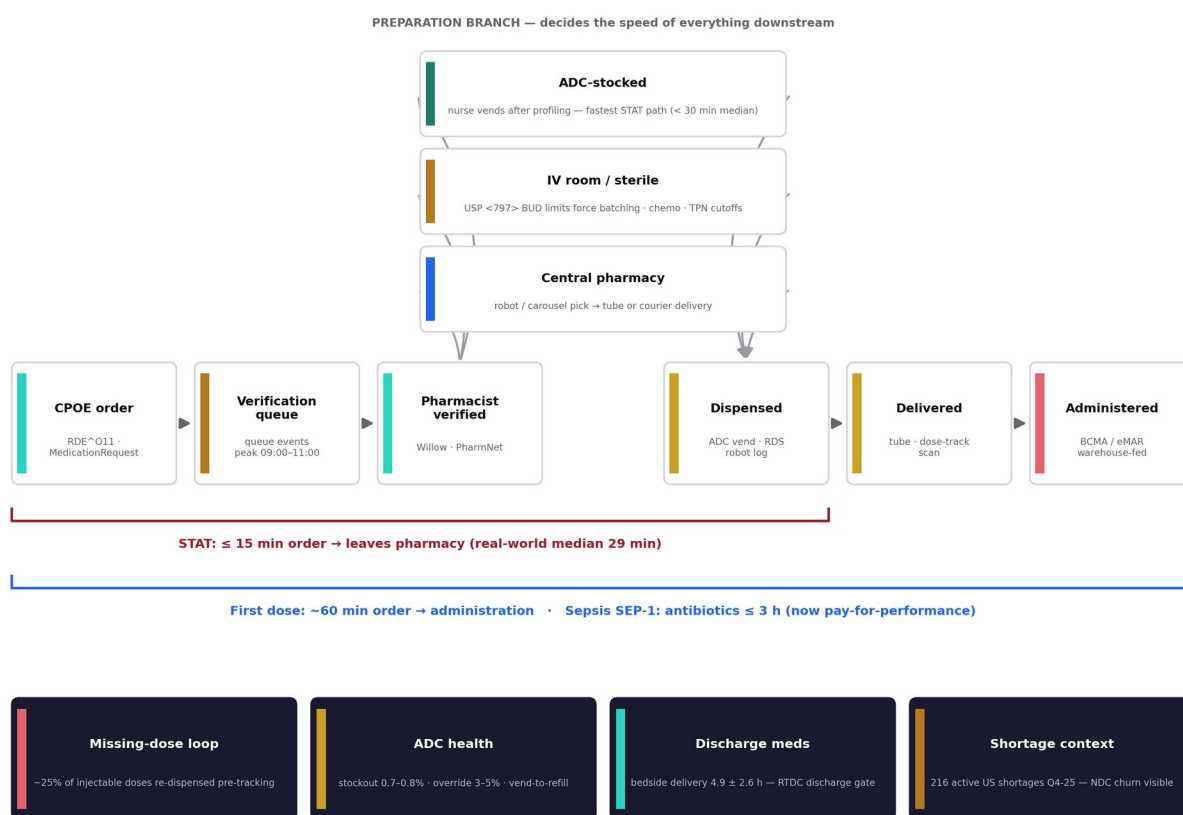


Figure 4 — The medication loop. The preparation branch — ADC-stocked versus IV room versus central pharmacy — decides the speed of everything downstream. Administration timestamps are warehouse-fed and honestly labeled.

9.1 KPI canon

KPI	Benchmark
Verification queue depth + verify TAT	No national benchmark — the telemetry itself differentiates; inflow peaks 09:00–11:00; 700–1,400 verifications per pharmacist shift
STAT order → dispense	≤ 15 min target; published real-world median 29 min

KPI	Benchmark
First dose order → administration	~60 min target; split at verification per the AHRQ measure
Time-to-antibiotic (sepsis)	SEP-1 three-hour window — now in Medicare pay-for-performance
Missing doses / re-request loops	~25% of injectable doses re-dispensed pre-tracking; QI programs reach ~1 per 100 doses
ADC stockout / override rates	Stockout ~0.7–0.8% at optimized sites; override targets 3–5% of transactions (ISMP-reviewed)
IV room batch adherence and waste	4.85% ICU CSP waste baseline; 1 → 3 batches/day cut waste 31% (~\$183K/yr)
Chemo preparation TAT	~90 min baseline; ~24–27 min with advance preparation
Discharge-med readiness	% planned discharges with meds ready by target; bedside waits measured at 4.9 ± 2.6 h
BCMA scan compliance	≥ 95% (leading organizations 98%)
Controlled-substance discrepancy aging	Resolve by shift end (ASHP); unresolved count is cockpit-grade

9.2 Barrier catalog

Enumerated barriers: the post-rounds verification surge and shift-boundary dips; the technician shortage (roughly 88% of hospitals short of experienced technicians, above 92% for sterile compounding); IV-room capacity bounded by USP <797> beyond-use dating; 503B outsourcing dependency (89% of hospitals rely exclusively on registered outsourcing facilities); drug shortages — 216 active in Q4 2025, three-quarters of them chronic since 2022, driving NDC churn and therapeutic substitution that is directly visible in flow data; the missing-dose loop; ADC par misconfiguration; tube-system downtime and delivery batching; prior authorization on discharge medications; controlled-substance workflow friction; and the structural tension between pharmacist stewardship time and queue time.

9.3 Screens

Route	Surface	What it shows
/pharmacy	Medication Flow Board (workspace)	Live verification queue (depth, age, STAT flags); first-dose clocks; sepsis antibiotic timers; breach chips
/pharmacy/discharge-meds	Discharge Med Readiness (workspace)	THE RTDC HERO: today's planned discharges × med status pipeline (not started / PA pending / filling / ready / delivered)
/pharmacy/iv-room	IV Room & Batches (workspace)	Batch cycle board with BUD windows; TPN cutoff countdown; chemo prep timeline; waste tracker
/pharmacy/dispense	Dispense & Delivery (workspace)	ADC stockout/override tiles by unit; missing-dose loop tracker; delivery segment timers
/pharmacy/controlled	Controlled Substances (workspace)	Open discrepancies with resolution aging; override outliers at unit/station level — analytics only, no individual diversion scoring

Route	Surface	What it shows
/analytics/pharmacy-tat	Pharmacy TAT Analytics (study)	Verify/dispense/admin waterfalls; queue-depth heatmap by hour × weekday; shortage impact view
Cockpit tile	Pharmacy health	Queue depth vs hour-norm; oldest STAT unverified; sepsis clocks at risk; stockouts on shortage drugs

9.4 Transactional systems and integration posture

System class	Market leaders (2025–26)	Zephyrus integration
EHR pharmacy	Epic Willow (43.7% hospitals / 56.9% beds) · Oracle PharmNet (shrinking) · MEDITECH Expanse	RDE/RDS real-time; verify-queue events via open.epic; Clarity MedicationAdministrationFact for BCMA
ADCs	BD Pyxis (Pyxis Pro + BD Incada cloud, Oct 2025) and Omnicell XT (Titan XT, Dec 2025) — a de facto duopoly	The ADC transaction feed (vend/refill/return/waste/override with user, drawer, timestamp) — the richest real-time pharmacy source
Central automation	Omnicell XR2 · Swisslog PillPick / BoxPicker · ARxIUM	Dispense-event logs where present
IV workflow	BD PharmacyKeeper · Omnicell IVX · Epic Dispense Prep (DoseEdge sunsets 2028 — do not build against it)	Prep milestones — optional (< 50% adoption); degrades to verify → dispense
Diversion analytics	Bluesight ControlCheck · BD HealthSight · Omnicell · Medacis	Out of scope to replicate; Zephyrus surfaces ops-level override and discrepancy signals only
Drug data	First Databank · Medi-Span · Micromedex · Multum	Normalize to RxNorm for rollups; keep NDC for shortage and inventory matching

TWO HONESTY RULES

Administration timestamps come from the nightly warehouse (US Core FHIR omits MedicationAdministration; the HL7 RAS message is rare in practice). Any metric that depends on them is labeled "as of last sync" and never presented as real-time.

Zephyrus surfaces override and discrepancy outliers at the unit and station level. Naming individuals as diversion risks is a regulated-adjacent product category (Bluesight, BD) and is explicitly out of scope.

PHARMACY SATELLITE TABLES

prod.rx_orders (RxNorm + NDC, priority class, IV/hazardous/controlled flags, shortage flag) · prod.rx_verifications (queue in/out, pharmacist) · prod.rx_preps (batch, BUD expiry, chemo/TPN flags) · prod.rx_dispenses (source: ADC/robot/carousel/manual) · prod.rx_administrations (warehouse-fed) · prod.adc_stations + adc_transactions · prod.rx_discharge_queue (med status pipeline vs discharge target) · hosp_ref.rx_formulary.

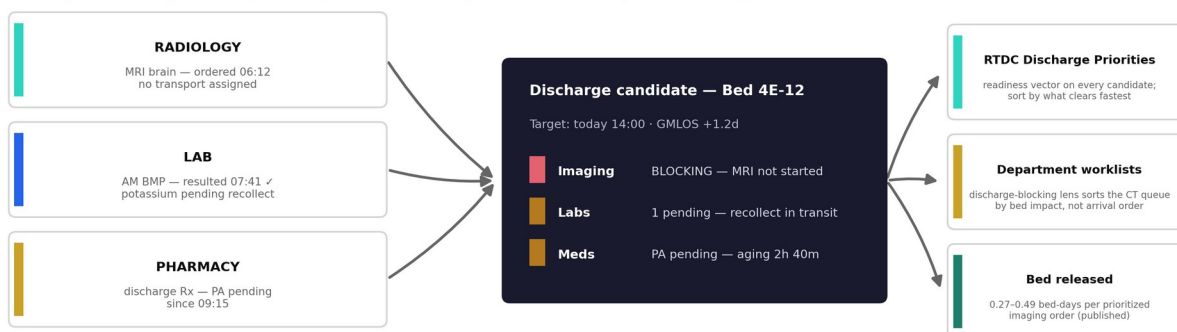
PART IV — CROSS-CUTTING & DELIVERY

10. The Discharge Readiness Vector

One feature stitches all three modules to the platform's core promise, and it deserves its own chapter. Every discharge candidate on the RTDC Discharge Priorities board gains a readiness vector — imaging status, lab status, medication status — computed live from the three milestone spines. One glance answers why a discharge is stuck; one click reaches the blocking order; and the department worklists reciprocate with a discharge-blocking lens that sorts their queues by bed impact rather than arrival order. This is the Qventus-proven value (0.27–0.49 bed-days per prioritized order) delivered through the surface Zephyrus users already trust, rather than through a separate nudge product.

The Discharge Readiness Vector

Three spines converge on every discharge candidate: one glance answers “why is this discharge stuck?”



Accretes across phases: imaging axis ships with Radiology (R-11), labs with Path/Lab (L-11), meds with Pharmacy (X-7) — each phase upgrades the discharge board.

The same joins power the ED tracking board (pending-ancillary chips per patient) and the Periop case timeline (T&S gate · frozen-section timer · pre-op antibiotics).

Figure 5 — The readiness vector joins three spines to one discharge candidate — and accretes one axis per delivery phase.

The vector deliberately accretes across the roadmap rather than shipping as a late integration phase: the imaging axis ships inside Phase 1 (task R-11), the lab axis inside Phase 2 (L-11), and the medication axis inside Phase 3 (X-7). Each phase therefore upgrades the RTDC discharge board on delivery day, and the cross-module joins are exercised continuously instead of deferred. The same encounter joins power the ED tracking board's pending-ancillary chips — the pattern the JACR literature credits with eliminating status-inquiry calls — and the Perioperative case timeline's gates: type-and-screen readiness, the intraoperative frozen-section timer, and pre-op antibiotic timing.

11. Predictions, Process Intelligence, and Benchmarks

Phase 4 adds a predictive layer consistent with the RTDC module's posture, one forecast per department, each surfaced as a sort order or a planning view rather than an alarm. For

radiology, TAT-breach risk scoring on open orders — the published state of the art reaches an AUC of 0.85 predicting large overnight delays — plus hourly scanner-demand forecasts feeding the RTDC demand model. For the lab, an AM-result readiness forecast answering the morning huddle's real question: will the potassium be back before rounds? For pharmacy, queue-depth forecasting by hour and weekday, stockout prediction from dispense velocity against par levels, and discharge-medication readiness ETAs.

Because every milestone is registered in the OCEL process-model landscape, the Study altitude's process-intelligence machinery — conformance and performance signals in the arena schema — operates on ancillary flows without new interfaces. And a benchmark reference registry seeds every Study chart with the published reference lines cited throughout this document, each carrying its source citation on hover: the CAP Q-Probes medians, the ASHP survey figures, the Joint Commission targets. Benchmarks from older-but-canonical studies are labeled "established benchmark," never passed off as current national data.

12. Phased Roadmap

Delivery runs nine to twelve sprints across five phases. Phase 0 builds the shared spine — migrations, models, the milestone projector and SLA evaluator, parser plumbing on the existing integration runtime, OCEL registration, demo scaffolding, and the reusable timeline component. Each department phase then follows the same rhythm: satellite tables and parsers first, the flow board and worklist second, the cross-module joins and cockpit tile third, the Study analytics last. Every phase ends with a verification gate: the full test suite, the UI-canon script, demo invariants, and a design audit.

Phased roadmap — nine to twelve sprints, each phase independently shippable

Demo data first: every screen works before any interface is connected. Feeds are enrichment, not prerequisites.



Readiness-vector accretion milestones (diamond markers): the RTDC discharge board gains one axis per phase.

Figure 6 — The delivery plan. Diamond markers show the readiness vector accreting one axis per phase.

Phase	Duration	Delivers
Phase 0 — Shared spine	1 sprint	Spine tables + models · milestone projector · SLA evaluator · parser plumbing · OCEL registration · demo scaffolding · timeline component · shared UI kit
Phase 1 — Radiology	2–3 sprints	ORM/ORU/MPPS parsers · flow board · worklist · modality utilization · reads & results · cockpit tile · ED chips + discharge imaging axis · TAT study · IR analytics
Phase 2 — Pathology & Lab	2–3 sprints	OML/ORU parsers · lab flow board · specimen tracker · decision-pending results · blood bank · AP aging · critical-value tracker · lab axis · TAT study
Phase 3 — Inpatient Pharmacy	2–3 sprints	RDE/RDS + queue-event parsers · ADC transactions · med flow board · discharge med readiness (vector complete) · IV room · dispense · controlled substances · TAT study
Phase 4 — Predictive + polish	1–2 sprints	Per-department forecasts · follow-up recommendation tracking · benchmark pack · coherence pass · accessibility and canon audits

DEMO-FIRST SEQUENCING

Every phase is demoable with zero interfaces connected: the demo generators produce coherent milestone spines, and feeds upgrade fidelity when they arrive. This keeps the sales narrative and the engineering sequence identical — the product is always showable.

13. Risks and Open Questions

13.1 Managed risks

Risk	Mitigation
Feed availability varies by site (MPPS, tube events, IVWMS, middleware are all optional)	Every screen defines a degraded-mode milestone set; worst case renders from ORM/ORU alone. The source-of-truth catalog drives what appears.
Administration data latency (no US Core MedicationAdministration; RAS rare)	Warehouse-fed metrics carry an "as of last sync" label enforced at the component level; never presented as real-time.
Alarm fatigue from three status-dense modules	Thresholds live in SLA definitions; cockpit shows counts and oldest-age only; earned-urgency review is a phase exit gate.
Scope creep into transactional territory ("acknowledge order," "reprint label")	Phases 1–3 are read-only plus barrier annotation. Write-back is a separate future decision.
Diversion-adjacent liability	Unit/station-level signals only; individual risk scoring is explicitly out of scope.
Benchmark hygiene	Reference registry stores citations; older canonical studies labeled "established benchmark."

13.2 Open questions

Four decisions are the author's to make before Phase 1 planning: which EHR the design-partner site runs (the Epic-first assumption is baked into feed priorities); whether interventional

radiology belongs to the Radiology module or to Perioperative ownership; whether outpatient imaging access analytics (no-shows, templates) ship in Phase 1 or defer, being the least connected to inpatient flow; and whether the demo narrative requires all three departments simultaneously or staged phase by phase.

APPENDICES

Appendix A — Source-of-Truth Milestone Maps

The ingestion contract per department: which system asserts each milestone, and through which feed. Where two systems assert the same milestone, both assertions are retained and the SLA engine prefers the listed source of truth.

A.1 Radiology

Milestone	Source of truth	Feed
Order placed / priority	EHR CPOE	ORM/OMI (ORC-1=NW) · FHIR ServiceRequest
Protocol assigned	RIS / EHR protocol activity	RIS event · Clarity backfill
Scheduled slot	RIS scheduling	SIU · FHIR Appointment
Patient arrival / class / unit	ADT	ADT A04 / A08 / A02
Transport request → complete	TeleTracking or Epic Rover	Vendor API · Clarity
Exam start / end	Modality via DICOM MPPS; fallback RIS Begin/End Exam	MPPS N-CREATE / N-SET · ORM status (ORC-5 IP → CM)
Images in PACS	PACS / image manager	Storage commitment · ImagingStudy
Preliminary / final report	Reporting system (PowerScribe) → RIS	ORU OBR-25 = P / F (OBR-22 clock)
Critical result notified / acknowledged	CTRM	Vendor feed

A.2 Pathology & Laboratory

Milestone	Source of truth	Feed
Order placed	EHR CPOE	ORM / OML
Collected	Bedside barcode scan	OBR-7 (in ORM/ORU)
In transit	Pneumatic tube system	Swisslog carrier events (optional)
Received / accessioned	LIS accessioning	OBR-14
In analysis / auto-verified	Middleware (Instrument Manager)	Middleware events (optional)
Resulted / verified / corrected	LIS	ORU R01 — OBX-11/OBR-25 with OBR-22/OBX-19 clocks; LOINC in OBX-3
Critical value notified	LIS callback documentation	LIS event / Clarity
AP stage transitions	AP-LIS	Status events · Clarity backfill
Blood bank T&S / crossmatch / issue	Blood bank system	HCLL / SafeTrace / Beaker BB events

A.3 Inpatient Pharmacy

Milestone	Source of truth	Feed
Order placed	EHR CPOE	RDE^O11 / OMP · FHIR MedicationRequest
Verification queue in / out	EHR pharmacy module	open.epic verify-queue events
Verified	Willow / PharmNet	Queue-removal event · RDE acknowledgment
Prep start / complete / checked	IV workflow system	IVWMS events (optional; < 50% adoption)
Dispensed	ADC / robot / carousel	ADC transaction feed (vend) · RDS^O13 · robot logs
Delivered	Tube system / dose tracking	Carrier events · dose-track scans (optional)
Administered	BCMA / eMAR	Clarity/Caboodle MedicationAdministrationFact (nightly; RAS where available)
Waste / return / override	ADC	ADC transaction feed

Coding systems: LOINC for laboratory observations (with local-code fallback); RxNorm for medication rollups with NDC retained for shortage and inventory matching; DICOM modality codes for imaging.

Appendix B — Claude Code Implementation Prompt

The complete implementation backlog — sixty dependency-ordered tasks with acceptance criteria (ten spine, fifteen radiology, fourteen laboratory, fourteen pharmacy, seven predictive/polish) — is generated by the prompt below. It is reproduced verbatim from the markdown plan of record (docs/ANCILLARY-EXPANSION-RADIOLOGY-PATHOLOGY-PHARMACY-PLAN-2026-07-11.md, §11) so this document is self-contained; to use it, copy the block into Claude Code from the Zephyrus repository root. The prompt instructs Claude Code to create todos only — no code is written in that pass — with the global conventions (Token Canon, read-only posture, demo-first, protected auth system) embedded in every task it creates.

```
You are working in the Zephyrus repository (Laravel 11 + React 18 + Inertia 2 + PostgreSQL multi-
schema). Your ONLY task right now is to create the complete implementation todo list for the
Ancillary Services Expansion (Radiology, Pathology/Lab, Inpatient Pharmacy) using your todo/task
tools. DO NOT write any code, migrations, or files in this pass — create todos only, exactly as
specified below, preserving the given IDs in todo titles, the given order, and the stated
dependencies. After creating all todos, print a summary count by phase and stop.

BEFORE creating todos, read these files to load context (do not summarize them back to me):
- docs/ANCILLARY-EXPANSION-RADIOLOGY-PATHOLOGY-PHARMACY-PLAN-2026-07-11.md (the plan this todo
list implements — the spec of record)
- CLAUDE.md, PRODUCT.md, DESIGN.md (Token Canon, Two-System Rule, earned urgency)
- resources/js/config/navigationConfig.ts (navigation SSOT — the ONLY place nav changes)
- app/Services/Rtdc/AncillaryServicesService.php and resources/js/Pages/RTDC/AncillaryServices.jsx
(the existing ancillary summary these modules deepen)
- app/Services/Rtdc/DischargePrioritiesService.php (extended by the Discharge Readiness Vector)
- database/migrations/2026_07_10_000200_operationalize_integration_runtime.php and
app/Models/Integration/*, app/Models/Raw/* (ingestion substrate)
- app/Services/Demo/OperationalDemoDataService.php, DemoClock.php, DistributionSampler.php,
DemoInvariantService.php (demo-data pattern)
- database/migrations/2026_07_10_000100_create_ocel_process_model_landscape_tables.php (OCEL
registration target)

GLOBAL CONVENTIONS — encode these as the description preamble of EVERY todo you create:
- Read-only analytics posture: no write-back to RIS/LIS/pharmacy systems. Barrier annotation writes
stay inside Zephyrus tables.
- New canonical tables live in the prod schema; reference catalogs in hosp_ref; ingestion artifacts
in integration/raw; process mining in ocel. Follow existing migration style (DB::statement CREATE
SCHEMA IF NOT EXISTS guard, dated filenames).
- Controllers are thin; each page's props come from a single service class with a build() method
(pattern: AncillaryServicesService). Services live in app/Services/Radiology|Lab|Pharmacy/.
- Pages live in resources/js/Pages/Radiology|Lab|Pharmacy/; shared components in
resources/js/Components/Ancillary/. Register routes in routes/web.php inside the authenticated
group with prefix/name groups matching existing style. Register navigation ONLY in
resources/js/config/navigationConfig.ts.
- Token Canon is a hard gate: healthcare-* tokens with dark: pairs only, no raw Tailwind palette,
no font-bold/text-[Npx] (cockpit components excepted per docs), tabular-nums on all metrics,
Card/Panel Surface primitives, PageContentLayout owns the gutter, status never by color alone,
earned-urgency thresholds come from ancillary_sla_definitions never hardcoded in JSX.
scripts/check-ui-canon.sh must pass.
- NEVER touch anything protected by .claude/rules/auth-system.md.
- Every screen must work in demo mode with no external feeds. Demo data flows through
OperationalDemoDataService + DemoClock + DistributionSampler and must satisfy DemoInvariantService
checks.
- Tests: Pest/PHPUnit feature tests for services and parsers; route-ownership tests derive from
navigationConfig; Vitest for React components with logic; every parser todo includes golden-message
fixture tests.
- Each todo below lists acceptance criteria (AC). A todo is not done until all its ACs pass.

CREATE THE FOLLOWING TODOS IN THIS ORDER.
```

== PHASE 0 — SHARED ANCILLARY SPINE ==

[P0-1] Migration: ancillary spine tables

Create migrations for: prod.ancillary_orders (id, department enum rad/lab/rx, encounter_id FK prod.encounters nullable, patient_class, priority, ordered_at, current_state, current_milestone_code, terminal_at, meta jsonb); prod.ancillary_milestones (order_id FK, milestone_code, occurred_at, source string, provenance_id nullable FK integration.provenance_records, meta jsonb; append-only – add trigger or DB grant pattern mirroring integration_configuration_audits append-only migration); prod.ancillary_sla_definitions (department, metric_key, label, start_milestone_code, stop_milestone_code, priority_class nullable, patient_class nullable, warning_minutes, breach_minutes, active); prod.ancillary_breaches (order_id, sla_id, opened_at, cleared_at nullable, barrier_id nullable FK to barriers), hosp_ref.ancillary_milestone_types (department, code, label, sequence, expected_source). Indexes: milestones (order_id, occurred_at); orders (department, current_state), (encounter_id); breaches partial index on cleared_at IS NULL.
AC: php artisan migrate fresh runs clean both directions; append-only enforcement verified by a feature test that expects UPDATE/DELETE on milestones to fail.

[P0-2] Eloquent models + factories for the spine
App\Models\Ancillary\{AncillaryOrder, AncillaryMilestone, AncillarySlaDefinition, AncillaryBreach}
with relationships (order->milestones ordered, order->breaches, breach->barrier, order->encounter), casts, and factories usable by demo + tests.
AC: model relationship tests pass; factories generate ordered milestone sequences.

[P0-3] Seed hosp_ref milestone catalogs + SLA definitions
Seeder loading the full milestone catalogs from the plan (\$4.1 RAD_*, \$5.1 LAB_*/AP_*/BB_*, \$6.1 RX_*) and the KPI/SLA rows from the plan's KPI tables (\$4.2, \$5.2, \$6.2) with warning/breach thresholds per priority and patient class (e.g., RAD stat order-final warning 90/breach 120 min; LAB troponin order-result warning 45/breach 60; RX stat order-dispense warning 10/breach 15; RX sepsis abx order-admin warning 150/breach 180). Thresholds are data, editable later – seed sensible defaults with a comment citing the plan section.
AC: seeder idempotent; every milestone code referenced by an SLA row exists in the catalog.

[P0-4] MilestoneProjector service
App\Services\Ancillary\MilestoneProjector consuming Integration\CanonicalEventRecord payloads: creates/updates AncillaryOrder, appends AncillaryMilestone with idempotency key (source + message control id + milestone code), advances current_state, tolerates out-of-order arrival (milestone occurred_at may precede current head), records provenance.
AC: unit tests for idempotent replay (same message twice → one milestone), out-of-order arrival, unknown-order auto-create, and conflicting duplicate milestone from two sources (both stored, source-of-truth flag set per hosp_ref.expected_source).

[P0-5] SlaEvaluator service + scheduler wiring
App\Services\Ancillary\SlaEvaluator: given an order, evaluate all applicable SLA definitions; open ancillary_breaches at breach threshold, clear on stop-milestone arrival; emit OperationalEvent on open/clear (reuse existing OperationalEvent model/patterns). Wire a per-minute scheduled command ancillary:evaluate-sl原因 plus event-driven evaluation from MilestoneProjector.
AC: time-travel tests (Carbon::setTestNow) prove open/clear lifecycle; no duplicate open breaches; OperationalEvents emitted exactly once per transition.

[P0-6] Integration source registrations + parser plumbing
Register integration.sources rows (via seeder/config) for: radiology.orm, radiology.oru, radiology.mpps, radiology.siu, lab.ohl, lab.oru, lab.middleware, pharmacy.rde, pharmacy.rds, pharmacy.verify_queue, pharmacy.adc_transactions, shared.adt. Create app\Services\Integrations\Parsers/ base contract: parse(RawInboundMessage): CanonicalEventRecord[] – pure, replayable through existing event replay jobs. No department parsers yet – just the contract, registration, and a NullParser wired end-to-end.
AC: a synthetic raw.inbound_messages row flows source → parser → canonical event → MilestoneProjector → prod.ancillary_milestones in one feature test.

[P0-7] OCEL registration for ancillary object/event types
Register object types imaging_order, specimen, ap_case, med_order and event types mirroring the milestone codes in the ocel.* landscape tables, aligned to the existing D1 (Lab) / D5 (Imaging) / D7 (Pathology) process families.
AC: process-intelligence views (arena conformance/performance signal queries) return rows for demo ancillary events without UI changes.

[P0-8] Demo generator scaffolding for the spine
Extend OperationalDemoDataService with an AncillaryDemoGenerator abstraction: per-department arrival processes (DistributionSampler), milestone progression with realistic stage distributions, configurable breach rate, DemoClock-driven so the wall display animates. Add DemoInvariantService invariants: strict milestone ordering per order, no orphans, breach/SLA consistency, cross-module referential coherence (demo discharge blockers must reference real demo ancillary orders).
AC: demo refresh produces coherent spines for all three departments (even before department modules exist); invariants pass.

[P0-9] AncillaryOrderTimeline React component
resources/js/Components/Ancillary/AncillaryOrderTimeline.jsx: horizontal milestone spine (done/current/pending), elapsed + SLA-remaining clocks in tabular-nums, earned-urgency coloring driven by SLA props (never hardcoded), degraded-mode rendering when optional milestones absent, accessible (labels + icons, not color alone).
AC: Vitest render tests for full, degraded, and breached states; check-ui-canon passes.

[P0-10] Ancillary shared UI kit
 resources/js/Components/Ancillary/: AgingHeatmap.jsx (rows × aging buckets with earned-urgency),
 SlaComplianceFile.jsx (percentile-based, definition-on-hover via existing MetricDefinition
 pattern), BarrierChip.jsx (links to Barrier records), QueueDepthSparkline.jsx.
 AC: all four render in a Storybook-style example page under Pages/Examples/ or existing examples
 surface; canon clean.

== PHASE 1 – RADIOLOGY ==

[R-1] Migration + models: radiology satellites
 prod.rad_exams, prod.rad_scanners, prod.rad_reads, prod.rad_critical_results,
 hosp_ref.rad_modalities, hosp_ref.rad_subspecialties per plan §4.5, FK to ancillary_orders. Models
 in App\Models\Radiology\
 AC: migrations reversible; factories produce modality-realistic exams (XR no protocol, CT/MRI with
 contrast/protocol flags, IR flagged procedural).

[R-2] HL7 parser: radiology ORM/OMI → milestones
 Parser mapping ORM: ORC-1=NW → RAD_ORDERED (priority from quantity/timing or ORC), ORC-5 status
 transitions → RAD_SCHEDULED / RAD_EXAM_START (IP) / RAD_EXAM_END (CM); cancellations terminal-state
 the order. Golden fixtures: STAT ED CT order, routine IP MRI, cancel, status-update sequence.
 AC: fixture tests green; replay idempotent; unparseable → raw.dead_letters with reason.

[R-3] HL7 parser: radiology ORU → read milestones
 ORU R01: OBR-25 P → RAD_PRELIM, F → RAD_FINAL, C → addendum (increments rad_reads.addendum_count,
 never regresses state); OBR-22 is the clock timestamp; populate rad_reads.
 AC: fixtures for prelim-then-final, final-only, corrected-after-final; TAT math verified against
 hand-computed deltas.

[R-4] DICOM MPPS ingestion → exam start/end
 Parser for MPPS N-CREATE (IN PROGRESS → RAD_EXAM_START) and N-SET (COMPLETED/DISCONTINUED →
 RAD_EXAM_END/terminal), arriving as forwarded events via the integration runtime (MPPS-to-HTTP
 relay assumed upstream; document this in the parser docblock). Where both MPPS and RIS assert
 start/end, MPPS wins per hosp_ref.expected_source.
 AC: conflict test proves MPPS precedence while both assertions retained.

[R-5] Radiology demo generator
 Per plan §3.3: modality mix, ED/IP/OP classes, ED CT order-complete median ~108 min IQR 57-182,
 night/weekend degradation, portable XR skipping transport, IR cases with procedural pattern,
 critical-result events, configurable breach rate.
 AC: generated distributions within tolerance of targets (statistical test with fixed seed);
 invariants pass.

[R-6] Service + page: Imaging Flow Board (/radiology)
 RadiologyFlowBoardService->build() + Pages/Radiology/FlowBoard.jsx: AgingHeatmap by modality ×
 aging bucket, breach chips, discharge-blocking lens (orders whose encounter is on the discharge
 priority list), barrier annotation drawer writing Barrier records. Route + controller
 (App\Http\Controllers\Radiology\
 AC: renders from demo data; barrier annotation persists and appears in Improvement views; canon
 clean.

[R-7] Service + page: Order Worklist (/radiology/worklist)
 Filterable list (modality/priority/class/state/breach) with AncillaryOrderTimeline per row
 expansion; transport segment timers when transport milestones present, hidden when absent (degraded
 mode).
 AC: filters server-driven via service; timeline states correct for full + degraded demo orders.

[R-8] Service + page: Modality Utilization (/radiology/modality)
 Per-scanner day view: exam blocks vs idle gaps, ED/IP/OP mix stacking, downtime windows from
 rad_scanners; utilization % with definition-on-hover (75% effective-utilization reference line).
 AC: idle-gap math tested; Nivo/consistent chart tokens; canon clean.

[R-9] Service + page: Reads & Results (/radiology/reads)
 Worklist depth by subspecialty, prelim-final aging, unread backlog growth sparkline, critical-
 results closed-loop tracker (notified-acked aging vs policy).
 AC: closed-loop states correct; no per-item alarm styling – counts + oldest-age only.

[R-10] Radiology cockpit tile
 Components/cockpit/RadiologyTile.jsx: oldest unread by priority, open breach count, scanner down
 count. Wall-density captions permitted here only.
 AC: tile registered in cockpit composition; degrades when no data.

[R-11] ED + RTDC context joins (imaging axis)
 (a) ED tracking board rows: pending-imaging chip with age vs SLA. (b) DischargePrioritiesService: add imaging axis of the readiness vector {pending_count, oldest_age, blocking_flag}.
 AC: feature tests prove a demo discharge candidate with pending CT shows blocked-on-imaging in RTDC and the chip in ED.

[R-12] Study page: TAT Analytics (/analytics/radiology-tat)
 Waterfall of segment medians/P90s (which segment dominates), trend lines by priority/modality/shift, night/weekend split, breach Pareto by barrier type. Percentiles only – never mean alone.
 AC: percentile math unit-tested; page registered under the Study/Analytics domain in navigationConfig.

[R-13] Study page: IR Suite Analytics (/analytics/ir-utilization)
 Clone Periop analytics (FCOTS, turnover, utilization) against rad_exams where is_ir, reusing existing Periop components/services where importable.
 AC: metrics match Periop implementations on identical fixture data.

[R-14] Navigation + route-ownership registration (Radiology)
 Add Radiology domain to navigationConfig.ts (workspace domain with dashboardHref /radiology; Study pages homed under Analytics), icons from lucide, matchPrefixes ['/radiology'], excludePrefixes ['/radiology/analytics'] if applicable; routes/web.php group; route-ownership tests updated.
 AC: nav renders on desktop + mobile + command palette; route-ownership test green.

[R-15] Radiology phase verification gate
 Run scripts/check-ui-canon.sh, full test suite, demo refresh + invariants, and an /impeccable audit pass on the four new pages; fix all findings.
 AC: all green; screenshot set saved to docs/screenshots/.

== PHASE 2 – PATHOLOGY & LAB ==

[L-1] Migration + models: lab satellites
 prod.lab_specimens, prod.lab_results, prod.lab_critical_values, prod.ap_cases, prod.bb_readiness, hosp_ref.lab_test_catalog (with decision_class enum ed_disposition/discharge_gate/or_gate/none) per plan \$5.5. Models in App\Models\Lab\
 AC: recollect loop representable (parent_specimen_id); frozen-section ap_case links or_case_id to existing ORCase.

[L-2] HL7 parser: lab OML/ORM – order/collection milestones
 LAB_ORDERED from order message; LAB_COLLECTED from OBR-7; nurse-collect vs lab-collect from order detail where present. Fixtures: ED stat troponin, routine AM BMP, add-on order.
 AC: fixture tests green; OBR-7 absence handled (collection asserted later by ORU).

[L-3] HL7 parser: lab ORU – receipt/result milestones
 OBR-14 → LAB_RECEIVED; OBX-11/OBR-25 P/F/C → LAB_PRELIM/LAB_FINAL/LAB_CORRECTED with OBR-22/OBX-19 clocks; auto-verified flag from meta where middleware asserts it; critical_flag from abnormal flags (OBX-8 HH/LL); LOINC from OBX-3 with local fallback; populate lab_results.
 AC: fixtures for stat-with-critical, hemolyzed-rejected (→ LAB_RECOLLECT + linked specimen), micro preliminary-then-final culture sequence.

[L-4] Lab demo generator
 AM-draw wave (03:00-06:00 surge), stat/routine mix tied to demo ED census, hemolysis/recollect loops ~2-8% on ED draws, auto-verification 50-90%, critical values with median-7-min callbacks, AP cases with overnight-batch stage timing, frozen sections tied to demo OR cases, blood-bank readiness against the OR schedule.
 AC: distribution tolerances met with fixed seed; invariants pass; frozen-section demo events reference real demo OR cases.

[L-5] Service + page: Lab Flow Board (/lab)
 Stage waterfall (collect→receive→result→verify→notify) live view, stat-compliance ration tiles (e.g., % troponin < 60 min), pre-analytic quality strip (hemolysis %, rejection % vs 0.31% benchmark, BCC vs ≤3%), barrier drawer.
 AC: renders from demo; benchmark reference lines carry definition-on-hover with source labels.

[L-6] Service + page: Specimen Tracker (/lab/specimens)
 Per-specimen AncillaryOrderTimeline, recollect-loop flagging (linked spines), tube-transport segment when feed present.
 AC: recollect pairs render linked; degraded mode without transport feed.

[L-7] Service + page: Decision-Pending Results (/lab/pending-decisions) – THE HERO
 Pending results joined to downstream decisions via lab_test_catalog.decision_class: gates an ED disposition (encounter in ED tracking), a discharge (encounter in discharge priorities), or an OR

start (bb/frozen linkage). Sort by downstream impact; each row shows what it is blocking and for how long.

AC: feature tests for all three decision classes; RTDC/ED joins verified against demo data.

[L-8] Service + page: Blood Bank Readiness (/lab/blood-bank)

T&S/crossmatch status matrix vs today's OR schedule; readiness gate flags on Periop case timeline (join into existing case timeline component); MTP activity indicator.

AC: an OR case without complete T&S by T-minus threshold shows gated in both views.

[L-9] Service + page: AP Case Aging (/lab/anatomic-path)

Case-aging board by stage (grossed/processing/slides/diagnosed/signed-out) vs CAP 2-day biopsy guidance; frozen-section live timer surfaced on the OR case during intraoperative window.

AC: aging buckets correct across DemoClock advancement; frozen timer appears only during live demo OR case.

[L-10] Lab critical-value SLA tracker + cockpit tile

Critical-value notification aging vs policy inside the Flow Board + Components/cockpit/LabTile.jsx (stat compliance now, oldest decision-pending item, critical-value SLA state).

AC: tile registered; counts-and-oldest-age only.

[L-11] Discharge readiness vector: lab axis + ED chips

Extend DischargePrioritiesService with labs axis; ED tracking rows gain pending-lab chip. Same AC pattern as R-11.

[L-12] Study page: Lab TAT Analytics (/analytics/lab-tat)

Percentile trends by test/priority/shift, AM-draw readiness curve (% resulted by 07/08/09:00), auto-verification trend, breach Pareto by barrier, collect-vs-analytic segment split.

AC: percentile + readiness-curve math unit-tested; registered in Study domain.

[L-13] Navigation + routes (Lab domain) – mirror R-14.

[L-14] Lab phase verification gate – mirror R-15.

=== PHASE 3 – INPATIENT PHARMACY ===

[X-1] Migration + models: pharmacy satellites

prod.rx_orders, rx_verifications, rx_preps, rx_dispenses, rx_administrations, adc_stations, adc_transactions, rx_discharge_queue, hosp_ref.rx_formulary per plan \$6.5 (RxNorm CUI + NDC columns; is_controlled/is_hazardous/shortage_flag). Models in App\Models\Pharmacy\.

AC: reversible; factories cover STAT/first-dose/IV-batch/chemo/TPN/discharge classes.

[X-2] HL7 parser: RDE/RDS + verify-queue events

RDE^011 → RX_ORDERED (priority from timing/priority fields); verify-queue add/remove events (open.epic Ancillary Pharmacy shape) → RX_QUEUE_IN/RX_VERIFIED; RDS^013 → RX_DISPENSED. Fixtures: STAT abx for sepsis-flagged encounter, routine first dose, order modify, discontinue.

AC: fixtures green; queue add/remove idempotent under replay; discontinue terminal-states the order.

[X-3] ADC transaction ingestion

Parser for Pyxis/Omniceil-style transaction feeds (CSV/HL7 DFT-ish; define a canonical adc_transactions shape and document vendor mapping in the docblock): vend → RX_DISPENSED (when linkable to an order) plus station-level rollups (stockout events, overrides, refill, waste/return).

AC: vend-with-order links milestone; vend-without-order (override) recorded station-level only; stockout/override rollup queries tested.

[X-4] Warehouse-fed administration ingestion

Batch importer contract for BCMA administration extracts (Clarity MedicationAdministrationFact shape) → RX_ADMINISTERED with honest freshness labeling (as_of timestamp stored; UI must render "as of <sync>" – enforce via component contract).

AC: importer idempotent on re-run; freshness propagates to any component showing admin-derived metrics.

[X-5] Pharmacy demo generator

Verification-queue inflow peaking 09:00-11:00 with shift-boundary dips, STAT ≤15-min clock class, first-dose 60-min class, sepsis abx clocks tied to demo ED sepsis flags, IV/TPN batch cycles with cutoffs, ADC-stocked vs pharmacy-prepared branches, missing-dose re-request loops, discharge-med statuses tied to demo discharge list, stockout/override events, shortage-flagged drugs.

AC: distribution tolerances with fixed seed; invariants incl. discharge-queue coherence pass.

[X-6] Service + page: Medication Flow Board (/pharmacy)

Live verification queue (depth, age distribution, STAT flags), first-dose clocks, sepsis antibiotic timers (order-admin vs 3h with honest admin-freshness handling: real-time to dispense, labeled

beyond), breach chips, barrier drawer.

AC: queue metrics tested; sepsis timer uses dispense milestone real-time and labels admin-dependent tail.

[X-7] Service + page: Discharge Med Readiness (/pharmacy/discharge-meds) – completes the vector Today's planned discharges × med status pipeline (not started / PA pending / filling / ready / delivered), ready-by-target %, aging on PA-pending. Extend DischargePrioritiesService with meds axis (vector now complete: imaging+labs+meds).

AC: full vector renders in RTDC discharge priorities; pipeline transitions tested.

[X-8] Service + page: IV Room & Batches (/pharmacy/iv-room)

Batch cycle board with BUD windows, TPN cutoff countdown (e.g., 11:00), chemo prep timeline, waste tracker vs batch frequency.

AC: cutoff countdown DemoClock-correct; degraded mode when no IVWMS feed (verify-dispense only) renders without prep columns.

[X-9] Service + page: Dispense & Delivery (/pharmacy/dispense)

ADC stockout/override tiles by unit (vend-to-refill, stockout % vs ~0.7-0.8% reference), missing-dose loop tracker (re-request chains), delivery segment timers where feed present.

AC: rollup queries indexed + tested at demo scale.

[X-10] Service + page: Controlled Substances (/pharmacy/controlled)

Open discrepancy list with resolution aging (shift-end policy clock), override outlier tiles at unit/station level. Analytics only – NO individual-diversion scoring; add a code comment and page caption stating diversion investigation is out of scope.

AC: no user-level risk scoring anywhere; aging clocks tested.

[X-11] Pharmacy cockpit tile + ED boarder lens

Components/cockpit/PharmacyTile.jsx (queue depth vs hour-norm, oldest STAT unverified, sepsis clocks at risk, stockouts on shortage-flagged drugs). ED boarding view: boarder med-delay lens (pending home meds/abx for boarded patients).

AC: tile registered; boarder lens joins demo boarding list correctly.

[X-12] Study page: Pharmacy TAT Analytics (/analytics/pharmacy-tat)

Verify/dispense/admin waterfalls by priority/shift/unit, queue-depth heatmap (hour × weekday), missing-dose Pareto, shortage-impact view (shortage-flagged drug dispense velocity + substitution visibility).

AC: heatmap math tested; registered in Study domain.

[X-13] Navigation + routes (Pharmacy domain) – mirror R-14.

[X-14] Pharmacy phase verification gate – mirror R-15.

== PHASE 4 – PREDICTIVE LAYER + POLISH ==

[P4-1] Radiology TAT-breach prediction service: features (hour, weekday, modality, priority, queue depth, staffing proxy) → breach-risk score on open orders; surface as sort option on worklist + flow board (never an alarm). AC: backtested on demo history; calibration test.

[P4-2] Lab AM-readiness forecast: per-encounter probability that decision-class labs result before 09:00; surface in Decision-Pending and RTDC morning huddle views. AC: calibration on demo data.

[P4-3] Pharmacy queue-depth forecast (hour×day) + stockout prediction (dispense velocity × par × shortage flag); surface on flow board and dispense page. AC: forecast beats naive baseline on demo backtest.

[P4-4] Radiology follow-up recommendation tracking: rad_followup_recommendations table, ingestion stub, completion-rate Study widget (28-77% published completion is the motivating stat). AC: end-to-end demo flow.

[P4-5] Benchmark pack: reference-line registry (hosp_ref.benchmark.references: metric_key, value, label, source_citation) rendered on Study charts with citation-on-hover; seed from plan \$4.2/\$5.2/\$6.2 tables. AC: every Study chart shows sourced reference lines.

[P4-6] Cross-module coherence pass: ancillary readiness vector consistent across RTDC/ED/Periop surfaces; AncillaryServices.jsx summary tiles gain drill-through links to the three new workspaces. AC: navigation from every summary tile lands on a filtered department view.

[P4-7] Final verification gate: full test suite, check-ui-canon, demo invariants, /impeccable audit on all new pages, WCAG 2.2 AA review (contrast, focus-visible gold, keyboard nav on worklists), route-ownership tests, DEVLOG entry docs/DEVLOG-ancillary-expansion.md. AC: all green.

DEPENDENCIES to encode: every P0 todo blocks all R/L/X todos. R-1..R-5 block R-6..R-13; R-14 blocks R-15. Same shape within L and X phases. R-11 blocks L-11 blocks X-7 (the readiness vector accretes in that order). Phase 4 blocked by R-15, L-14, X-14.

Now create all todos with these IDs, titles, descriptions (including the global-conventions preamble and ACs), and dependencies. Then print the per-phase counts and stop.

Appendix C — Research Provenance

The findings in this document synthesize three parallel research streams (radiology, pathology/laboratory, inpatient pharmacy) conducted July 11, 2026, drawing on: peer-reviewed throughput and TAT studies (Annals of Emergency Medicine, AJR, JACR, Academic Radiology, BMC Emergency Medicine, JCO Oncology Practice); CAP Q-Probes and Q-Tracks benchmarking programs; ASHP national surveys and drug-shortage statistics; Joint Commission National Patient Safety Goals and Quick Safety publications; CMS rulemaking on the AUC program and SEP-1; KLAS market-share reporting (2026 report on 2025 data); vendor documentation (open.epic, BD, Omnicell, Swisslog, Data Innovations, Nuance, Sectra, TeleTracking); and the published results of the GE HealthCare Command Center, Qventus, and LeanTaaS programs. Inline source URLs for every figure cited here are preserved in the markdown plan of record in the repository, which should be treated as the citation index for this document.

PLAN OF RECORD

docs/ANCILLARY-EXPANSION-RADIOLOGY-PATHOLOGY-PHARMACY-PLAN-2026-07-11.md — the engineering-facing markdown version of this plan, including the full citation trail and the machine-readable implementation prompt.